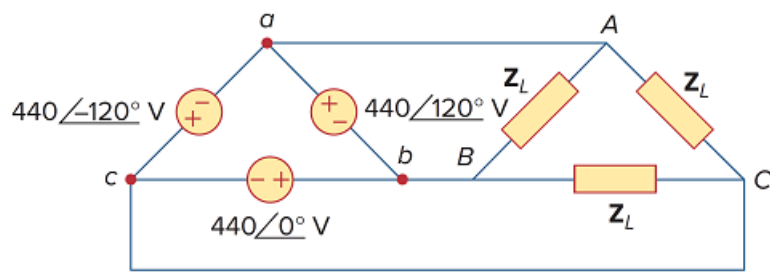


Problem

Three 440-V generators form a delta-connected source that is connected to a balanced delta-connected load of $Z_L = (8.66 + j5) \Omega$ per phase as shown in Fig. 12.52. Determine the value of I_{BC} and I_{aA} . What is the pf of the load?

Figure 12.52



Step-by-step solution

Step 1 of 5

Refer to Figure 12.52 in the text book.

Calculate the value of the current I_{BC} in the branch B-C.

$$I_{BC} = \frac{V_{BC}}{Z_L}$$

Substitute $440\angle 0^\circ \text{ V}$ for V_{BC} and $8.66 + j5 \Omega$ for Z_L .

$$\begin{aligned} I_{BC} &= \frac{440\angle 0^\circ}{8.66 + j5} \\ &= \frac{440\angle 0^\circ}{10\angle 30^\circ} \\ &= 44\angle -30^\circ \text{ A} \end{aligned}$$

Therefore, the value of the phase current I_{BC} is $44\angle -30^\circ \text{ A}$.

Step 2 of 5

Calculate the value of the current I_{AB} in the branch A-B.

$$I_{AB} = \frac{V_{AB}}{Z_L}$$

Substitute $440\angle 120^\circ \text{ V}$ for V_{AB} and $8.66 + j5 \Omega$ for Z_L .

$$\begin{aligned} I_{AB} &= \frac{440\angle 120^\circ}{8.66 + j5} \\ &= \frac{440\angle 120^\circ}{10\angle 30^\circ} \\ &= 44\angle 90^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_{AB} is $44\angle 90^\circ \text{ A}$.

Step 3 of 5

Calculate the value of the current I_{AC} in the branch A-C.

$$I_{AC} = \frac{V_{AC}}{Z_L}$$

Substitute $440\angle -120^\circ$ V for V_{AC} and $8.66 + j5 \Omega$ for Z_L .

$$\begin{aligned} I_{AC} &= \frac{440\angle -120^\circ}{8.66 + j5} \\ &= \frac{440\angle -120^\circ}{10\angle 30^\circ} \\ &= 44\angle -150^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_{AC} is $44\angle -150^\circ$ A.

Step 4 of 5

Calculate the value of the line current I_{aA} .

$$I_{aA} = I_{AB} - I_{AC}$$

Substitute $44\angle 90^\circ$ A for I_{AB} and $44\angle -150^\circ$ A for I_{AC} .

$$\begin{aligned} I_{aA} &= 44\angle 90^\circ - 44\angle -150^\circ \\ &= 76.21\angle 60^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_{aA} is $76.21\angle 60^\circ$ A.

Step 5 of 5

Notice that the value of the phase angle θ_i of the phase current I_{AB} is, 90° . Also, the value of the phase angle θ_v of the voltage V_{AB} is 120° .

Calculate the value of the power factor of the load.

$$\text{pf} = \cos(\theta_v - \theta_i)$$

Substitute 90° for θ_i and 120° for θ_v .

$$\begin{aligned} \text{pf} &= \cos(120^\circ - 90^\circ) \\ &= \cos(30^\circ) \\ &= 0.866 \end{aligned}$$

Therefore, the value of the power factor of the load is 0.866 .